Deep Patel

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PROFESSIONAL SUMMARY

High school senior focused on aerospace engineering and rocket propulsion. Creator of ORBIX, an aerospace engineering web app, and designer of a 3D-printed rocket bell nozzle studied with CFD simulation. Dedicated youth leader and volunteer looking for opportunities to learn more about physics, engineering, and space science through hands-on projects and research.

EDUCATION

Chelmsford High School, Chelmsford, MA Expected Graduation: June, 2027

GPA: 4.1 (Weighted)

Relevant Courses: AP Physics C, AP Computer Science, Calculus, Advanced Engineering Principles & 3D Design

SKILLS AND INTERESTS

Technical Skills: Java, CAD (Fusion 360, Revit, Onshape, OpenSCAD), CFD Simulation (SimFlow), Robotics, Microsoft Office Suite, Mathematical Modeling, Adobe Creative Cloud

Interests: Aerospace Engineering, Mechanical Engineering, Astrophysics, Rocketry, 3D Modeling

TECHNICAL PROJECTS

Aerodynamic Bell Nozzle Sep 2025 - Present

Lead Designer & Fabricator (Self-Directed)

- Studied nozzle thermodynamics and isentropic flow independently to size a LOX/RP-1 bell nozzle for a 100 bar chamber expanding to 0.7 bar.
- Modeled the nozzle contour in OpenSCAD and checked its 15.92 area ratio against the Mach 3.6 isentropic design point.
- Set up and debugged a compressible-flow CFD case in SimFlow, tracing solver divergence through mesh, boundary condition, and limiter changes.
- 3D-printed the 84 mm nozzle in PLA, printing it inverted so no support material formed inside the bell.

ORBIX Web Application Aug 2026 - Present

Creator & Project Lead | https://orbix-aerospace.vercel.app/

- Conceived and engineered a full-stack web app with 13 engineering calculators, a mission planner, and profiles of 10 real aircraft and launch vehicles.
- Wrote the specifications for each tool, including equations, inputs, and units, and programmed the front-end logic and physics engines.
- Planned a verification page that checks results against published references (NACA Report 1135, U.S. Standard Atmosphere 1976); 26 of 36 values match within rounding.

ACADEMICS AND EXTRACURRICULARS

Track and Field Team Nov 2023 - Present

Student Athlete

- 3x MSTCA Div. II State Relay Medalist (High Jump: 4th, 4th, 6th) and 6-season varsity athlete (Indoor & Outdoor).
- Compete in high jump, 55m hurdles, and 110m hurdles. Lead daily workouts for 75+ athletes and mentor 35+ underclassmen on form and discipline.

Artificial Intelligence (AI) Club Sep 2025 - Present

Vice President

- Manage 15 members and host bi-weekly meetings focused on safe AI usage and coding ethics in the real world.
- Directed a project team for the MIT URTC competition, building a custom AI software tool that monitors home environments to optimize care for household plants.

Robotics Club Sep 2023 - Jun 2024

Member

- Worked with a team to design, build, and test robots for VEX Robotics competitions.
- Programmed the robot to navigate competition mazes autonomously, and engineered the physical mechanisms needed to lift plastic balls into scoring baskets.

National Honor Society (NHS) Mar 2026 - Present

Member & Volunteer

WORK EXPERIENCE

ALPHAPEP LLC, Lowell MA May 2026 - Present

Email Marketing Associate

- Design and distribute targeted marketing emails to advertise a diverse variety of the company's products.
- Manage and execute outgoing email campaigns to ensure successful delivery of promotional materials and drive customer engagement.

LEADERSHIP AND VOLUNTEERING

BAPS Swaminarayan Sanstha (Bal Mandal & Kishore Mandal), Lowell MA Aug 2022 - Present

Youth Activities Coordinator & Volunteer *Aug 2024 - Present*
Teacher *Aug 2022 - Aug 2024*

- Lead weekly youth assemblies for 70+ Pre-K–8th grade children and manage a team of 15 high school peer teachers.
- Plan and teach weekly lessons covering cultural values, history, and leadership skills.
- Serve as a logistics volunteer for the high school teen wing (Kishore Mandal), helping coordinate stage setup and crowd control for major cultural festivals serving over 1,000 attendees.

BAPS Charities, Lowell MA Jun. 2017 - Present
Walkathon Organizer & Outreach Volunteer

- Organized the annual 'In the Joy of Others' Walk-Run for 650+ participants, setting up routes and managing event registration.
- Helped raise over $15,000 across high school years to directly benefit local municipal partners, including the Lowell Police Dept., Tyngsborough Fire Dept., and local public schools.

BAPS Akshardham, Robbinsville, NJ Jul. 2023 - Oct. 2023
Full-Time Volunteer

- Designed detailed architectural pieces using CorelDRAW and Adobe Illustrator, and created CNC machine files for a large cultural building project.
- Ran industrial CNC machines and hot wire cutters to make complex parts out of wood, PVC, and foam at scale.
- Worked with senior designers and other volunteers to ensure designs were accurate and finished on time without wasting materials.